Create Role Based Access Control

យើងដាក់លក្ខណអោយដោយប្រើប្រាស់ Role ដើម្បីកំណត់សិទ្ធអោយចូលប្រើ Resource Server យើងបានឬអត់

1- Create Authentication Request from Http

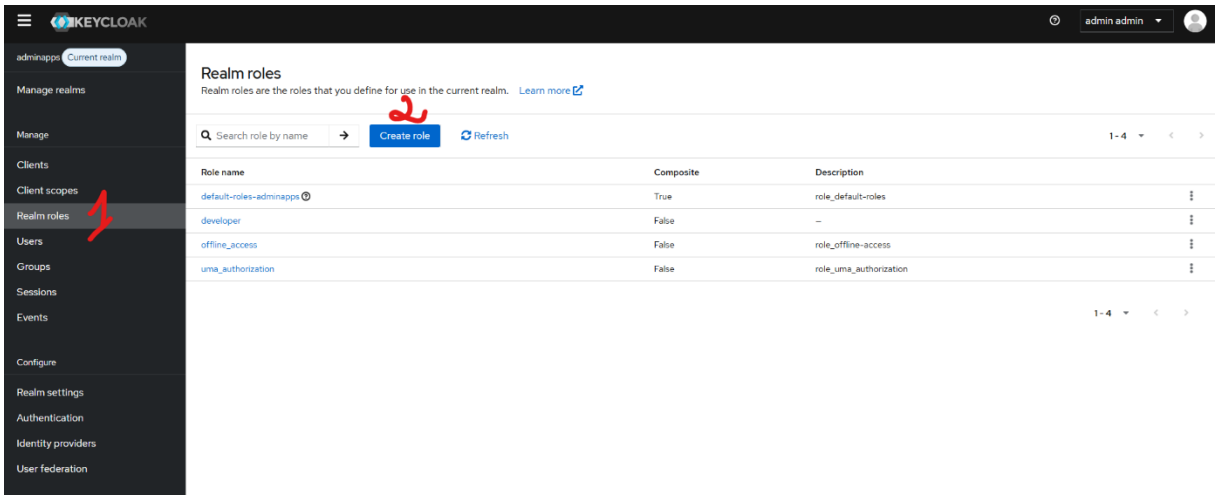
```
@Bean
SecurityFilterChain configure(HttpSecurity http) throws Exception{

    JwtAuthenticationConverter jwtAuthenticationConverter = new JwtAuthenticationConverter();
    jwtAuthenticationConverter.setJwtGrantedAuthoritiesConverter(new KeyCloakConverter());

    // Create object with filter chain
    http.authorizeHttpRequests(authz ->
                    authz
                            .requestMatchers(HttpMethod.GET, "/users/status/check")
                              .hasAuthority("SCOPE_profile")
                            .hasRole("developer1")
                              .hasAnyRole("USER","ADMIN")
                            .anyRequest().authenticated())
            .oauth2ResourceServer(oauth2 ->
                    oauth2.jwt(jwt ->
                            jwt.jwtAuthenticationConverter(jwtAuthenticationConverter)));
    return http.build();
}
```

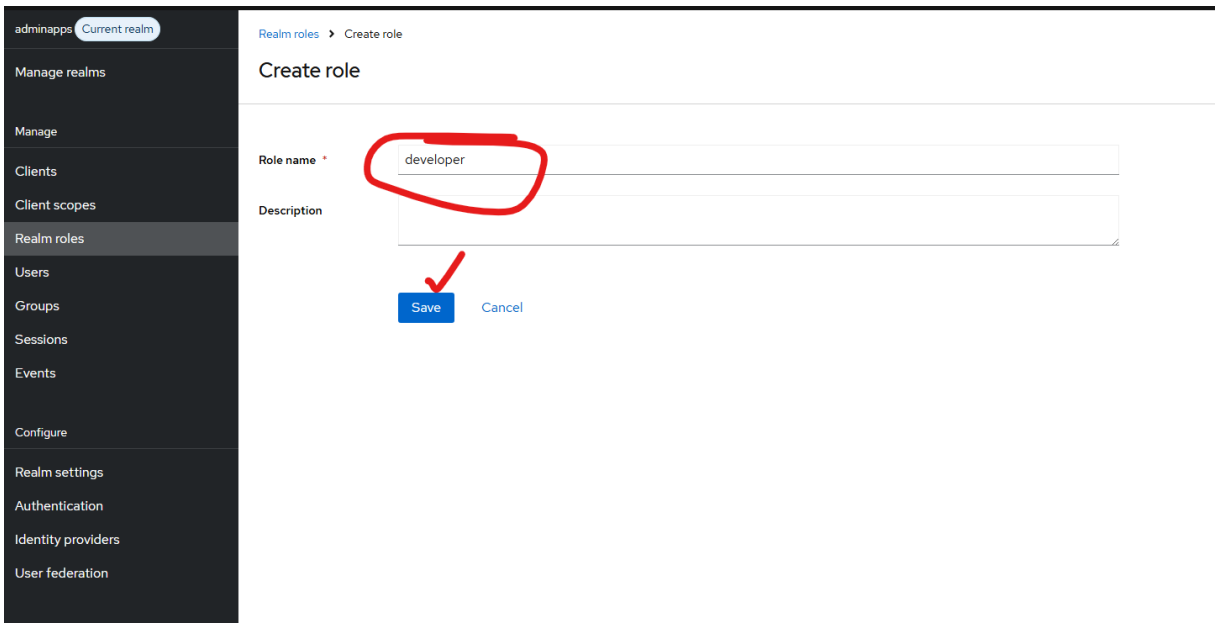
.hasRole("developer1") កន្លែងដែលយើងកំណត់ Role អាចធ្វើការ Access to the resource server

2- Create converter from JWT to get only the role

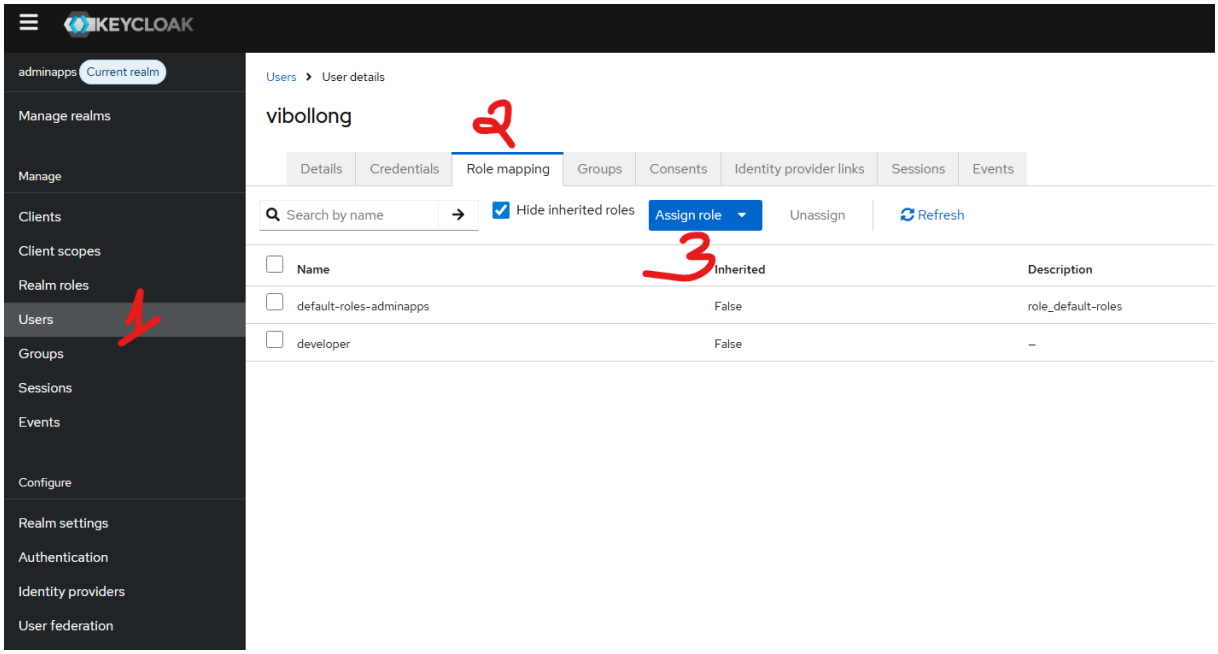
```
public class KeyCloakConverter implements Converter<Jwt, Collection<GrantedAuthority>> {
    @Override
    public Collection<GrantedAuthority> convert(Jwt jwt) {
        // this class will convert the Jwt object to the GrantedAuthority object
        Map<String, Object> realmAccess = (Map<String, Object>) jwt.getClaims().get("realm_access");

        if (realmAccess == null || realmAccess.isEmpty()) {
            return new ArrayList<>();
        }

        Collection<GrantedAuthority> returnValue = ((List<String>) realmAccess.get("roles"))
                .stream().map(roleName -> "ROLE_" + roleName)
                .map(SimpleGrantedAuthority::new)
                .collect(Collectors.toList());
        return returnValue;
    }
}
```

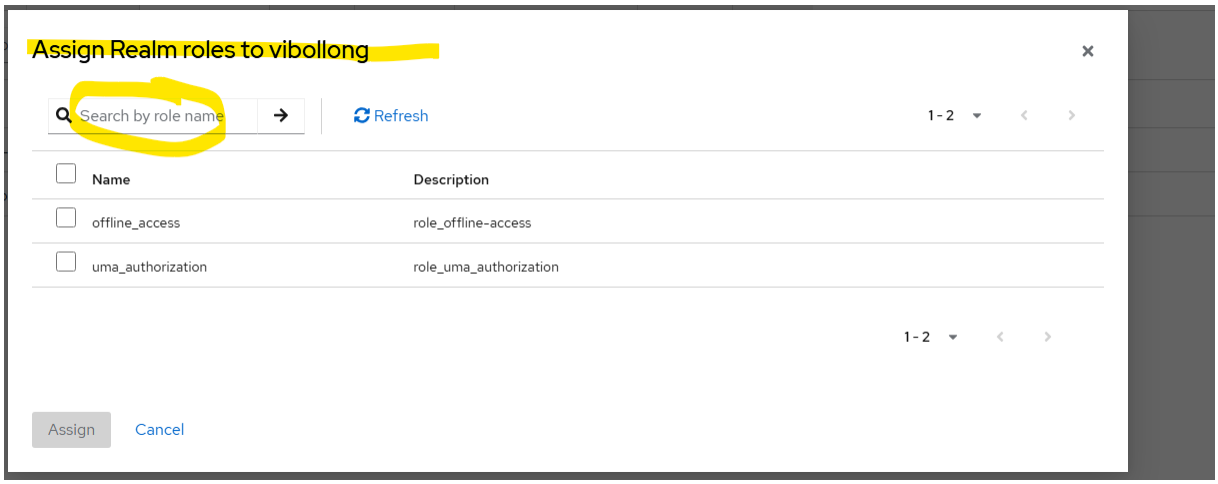
3- In the keyloak

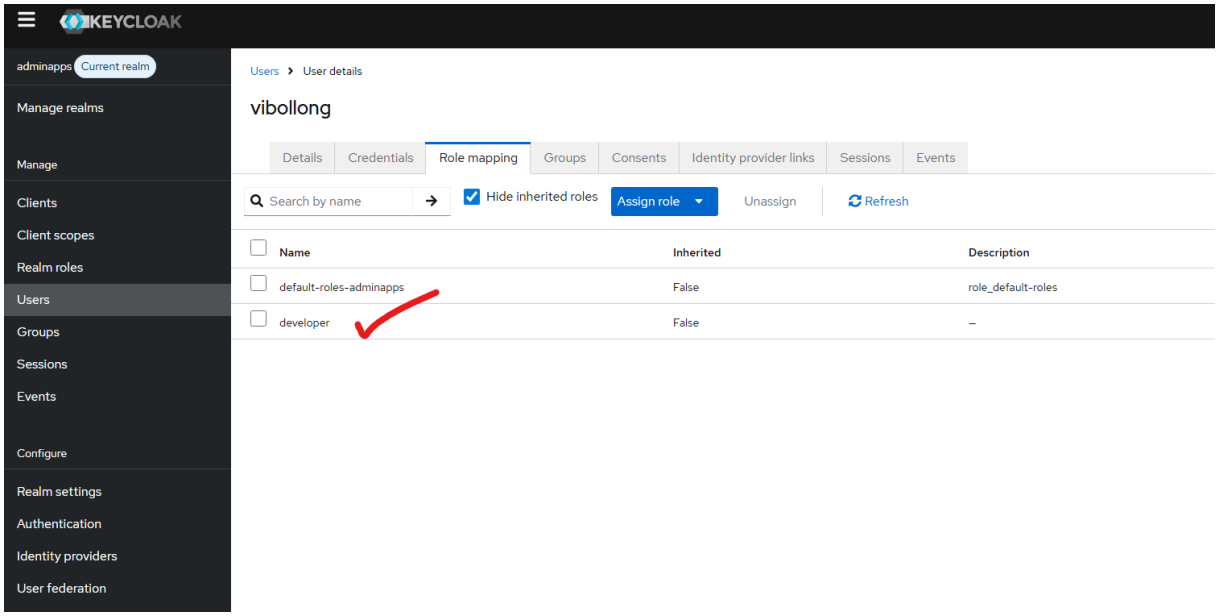


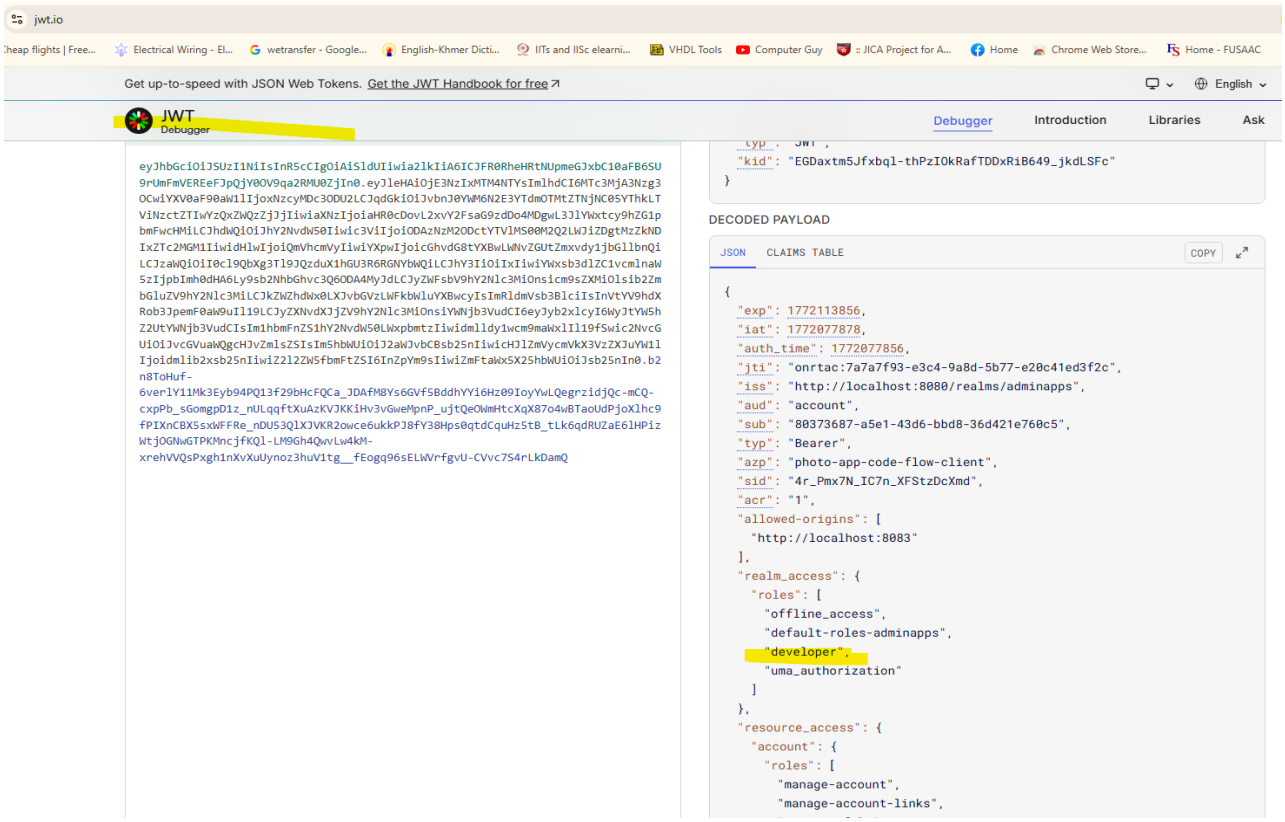


Assign Role to the user




Assign Realm roles to vibollong
Search by role name
Refresh
1 - 2
Name
Description
offline_access
role_offline-access
uma_authorization
role_uma_authorization
1 - 2
Assign
Cancel


KEYCLOAK
adminapps
Current realm
Manage realms
Manage
Clients
Client scopes
Realm roles
Users
Groups
Sessions
Events
Configure
Realm settings
Authentication
Identity providers
User federation
Users
User details
vibollong
Details
Credentials
Role mapping
Groups
Consents
Identity provider links
Sessions
Events
Search by name
Hide inherited roles
Assign role
Unassign
Refresh
Name
Inherited
Description
default-roles-adminapps
False
role_default-roles
developer
False
–


jwt.io
Get up-to-speed with JSON Web Tokens. Get the JWT Handbook for free
English
JWT Debugger
Debugger
Introduction
Libraries
Ask
"kid": "EGDaxtm5Jfxbql-thPzIOkRafTDDxRiB649_jkdLSFc"
DECODED PAYLOAD
JSON
CLAIMS TABLE
COPY
"exp": 1772113856,
"iat": 1772077878,
"auth_time": 1772077856,
"jti": "onrtac:7a7a7f93-e3c4-9a8d-5b77-e20c41ed3f2c",
"iss": "http://localhost:8080/realms/adminapps",
"aud": "account",
"sub": "80373687-a5e1-43d6-bbd8-36d421e760c5",
"typ": "Bearer",
"azp": "photo-app-code-flow-client",
"sid": "4r_Pmx7N_IC7n_XFStzDcXmd",
"acr": "1",
"allowed-origins": [
"http://localhost:8083"
],
"realm_access": {
"roles": [
"offline_access",
"default-roles-adminapps",
"developer",
"uma_authorization"
]
},
"resource_access": {
"account": {
"roles": [
"manage-account",
"manage-account-links",